

Syllabus

Instructor:	Dr. Randall R. Rojas
Office:	Bunche 8248
Office Hours:	Mon & Wed 3:30-4:30PM (via Zoom)
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Time and Location

Lec 80: Monday and Wednesday 11:00AM-12:15PM (Bunche 2221E)

Course Description

This course offers a rigorous and practical introduction to econometric methods at the graduate level, with a strong emphasis on implementation using Python and mathematical derivations. Students will build a solid foundation in classical linear regression, diagnostic testing, and model specification, progressing to more advanced topics such as panel data analysis, time series econometrics, and causal inference. Through applied coding exercises and real-world data analysis, students will learn to formulate, estimate, and interpret econometric models for academic research and policy evaluation.

Student Learning Outcomes (SLOs)

1. Derive and interpret classical and modern econometric models.
2. Implement econometric techniques using Python and relevant data analysis libraries.
3. Diagnose and address violations of model assumptions.
4. Evaluate causal relationships using quasi-experimental methods.
5. Apply panel and time series methods to economic data.
6. Critically assess econometric findings in academic literature.

Computation of Course Grade

- 30% Group Projects (Three projects, 10% each)
- 15% Homework (Three Assignments, 5% each)
- 20% Midterm Exam (Week 5)
- 35% Final Exam. Cumulative (Monday, Dec 8)
- 2% Extra Credit: Guest Speaker Lecture on October 13, 2025

Textbooks (Recommended)

1. Statistics (for reference if needed)

- Probability and Statistics: The Science of Uncertainty (2nd Ed.). Michael J. Evans & Jeffrey S. Rosenthal
- Probability and Statistics (4th Ed.). Morris H. DeGroot & Mark J. Schervish

2. Econometrics

- ^aPrinciples of Econometrics. Hill, R. C., Griffiths, W. E. and G. C. Lim, 5th Edition, 2018. Wiley and Sons.
- ^bUsing Python for Introductory Econometrics. Florian Heiss & Daniel Brunner (2020)
- Introductory Econometrics: A Modern Approach. Wooldridge, J., 7th Edition, 2020.
- Principles of Econometrics with R. C., Colonescu, 2016. <https://bookdown.org/ccolonescu/RPoE4/>
- Introduction to Python for Econometrics, Statistics and Data Analysis. K. Sheppard A Nielsen. O'Reilly (2020)

3. Programming (for reference if needed)

- Python for Data Analysis. Wes McKinney. O'Reilly (2nd Ed.)

Prerequisites

1. Python Bootcamp

2. Mathematics Bootcamp

- ### 3. Undergraduate level Econometrics course.
- Familiarity with a data analysis software (e.g., R, Excel, Matlab, STATA, etc.) and/or programming experience. For this courses, however, we will be using R and Python exclusively. In addition, you will be required to know LaTeX, Jupyter Notebook and R Markdown for preparation of your assignments.

Course Enforced Policies

- The use of electronic devices such as cellphones, tablets, and devices that allow you to communicate with others, is strictly banned. You will be reported for cheating if caught using them during exams. Therefore, please turn off all your electronic devices during exams.
- There are no extra credit assignments available.
- Scientific/graphing calculators are allowed and recommended during exams.
- All grades are final when filed by the instructor on the Final Grade Report.

Assignment Enforced Policies

- Late homework/project assignments will be deducted 1 point per minute late. In addition, if the submitted file is not in PDF or HTML format, there will be a 20 points deduction.
- Homework and project answers must be provided in the order in which they are asked on the respective assignment. If they are not in order, you will be deducted 20 points.
- Homework and projects must be typed using LaTeX, Jupyter Notebook, and/or R Markdown.
- There are no make up projects or homework assignments.
- Both projects and homework solutions must be submitted with the respective R and/or Python code. Therefore, incomplete project and/or homework solutions (e.g., missing the respective R code), will not receive any credit.
- You are advised to start working on the assignments as early as possible in order to have enough time to troubleshoot any coding issue(s) you may encounter.
- Although I encourage you to work together on the homework assignments, you must submit your own solution for each one to receive credit. Therefore, your homework assignment codes must be your own work, and cannot be plagiarized, otherwise you will not receive any credit.

Disabled Students and the Center for Accessible Education (CAE)

Any student with a preexisting illness or condition who requests special arrangements must (a) qualify under CAE rules for such special arrangements and (b) must take the exam with CAE. Any such arrangements with CAE must be made the first week of classes. The instructor must be informed of any such arrangement in the first week of classes. For additional information and the qualification conditions of the Center for Accessible Education, please visit their website at <http://www.cae.ucla.edu/>. All other students must take the exam at the scheduled time under the same time constraints. It is the responsibility of all students who request special arrangements with CAE to be familiar with all of their rules as well as the rules of this class.

Academic Dishonesty

Any cases of academic dishonesty will be reported to the Office of the Dean of Students. For more details please refer to the Office of the Dean of Students website at <http://www.studentgroups.ucla.edu/dos/>

Tentative Course Schedule

Week	Lecture Topics	Chapters
1 (Sep 29, Oct 1)	Classical Linear Regression Model	2 ^a , 3 ^b
2 (Oct 6, 8)	Prediction, Goodness-of-Fit, and Modeling Issues	4 ^a
3 (Oct 13, 15)	Multiple Regression Model Homework 1 (Due on Oct 15)	5 ^a
4 (Oct 20, 22)	Inference in the Multiple Regression Model Project 1 (Due on Oct 22)	6 ^a
5 (Oct 27, 29)	Causal Inference Midterm (TBD)	7 ^a
6 (Nov 3, 5)	Generalized and Advanced Estimation Techniques	8 ^a
7 (Nov 10, 12)	Time Series Econometrics Homework 2 (Due on Nov 12)	9 ^a
8 (Nov 17, 19)	Qualitative and Limited Dependent Variable Models Project 2 (Due on Nov 19)	16 ^a
9 (Nov 24, 26)	Endogeneity and Instrumental Variables Homework 3 (Due on Nov 26)	10 ^a , 11 ^a
10 (Dec 1, 3)	Panel Data Econometrics Project 3 (Due on Dec 3)	15 ^a
11 (December 8)	Final Exam (Monday, Dec 8)	Cumulative